Summary

The analysis draws on five interconnected datasets: *Churn Status*, *Customer Demographics*, *Customer Service*, *Transaction History*, and *Online Activity*. Each dataset contributes a different aspect of customer behaviour, allowing a fuller understanding of the drivers behind customer churn. At the heart of the study sits the *Churn Status* table, which provides the binary outcome variable indicating whether each customer has churned or remained active. All further analysis links back to this table, making it the logical starting point for exploring patterns and building explanatory insights.

The *Customer Demographics* dataset is included to understand how characteristics such as age, gender, income level and marital status relate to customer retention. These demographic attributes help identify which customer groups are more at risk of leaving, and whether certain segments behave differently under similar service conditions. This dataset is essential for identifying churn patterns across population segments and supporting targeted retention strategies.

The *Customer Service* dataset is incorporated because support interactions are often a direct predictor of churn. Variables such as interaction type, resolution status, and recency of service contact reveal a customer's satisfaction level and the handling of their issues. The final resolution outcome, whether resolved or unresolved, is particularly meaningful; customers whose last interaction is negative often display significantly higher churn rates. Including this dataset allows the analysis to link operational service performance to churn behaviour.

The *Transaction History* dataset provides behavioural signals relating to how frequently customers buy, when they last made a purchase, and how valuable they are to the organisation. These transactional indicators are commonly used in RFM (recency, frequency, monetary) analyses and directly align with churn prediction, as declining spending behaviour often precedes churn.

Finally, the *Online Activity* dataset offers additional behavioural context by capturing customers' digital engagement. Metrics such as last login date and activity frequency help identify customers who are becoming disengaged. Declining online engagement is often an early warning sign of potential churn, especially in digitally driven services.

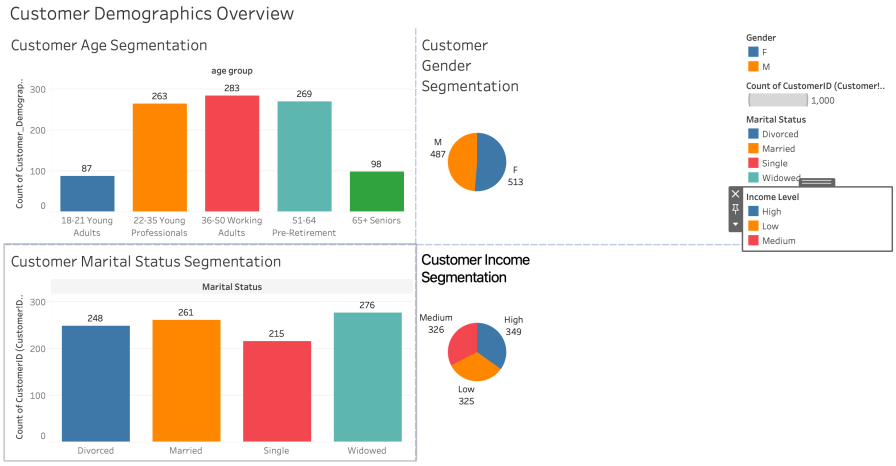
Together, these datasets give a multidimensional view of churn, combining outcomes, demographics, service experiences, financial behaviour and online engagement. This integrated structure ensures that the analysis captures not only who is churning, but why, enabling more accurate detection of risk patterns and more informed recommendations for retention.

Visualisations and statistical summaries

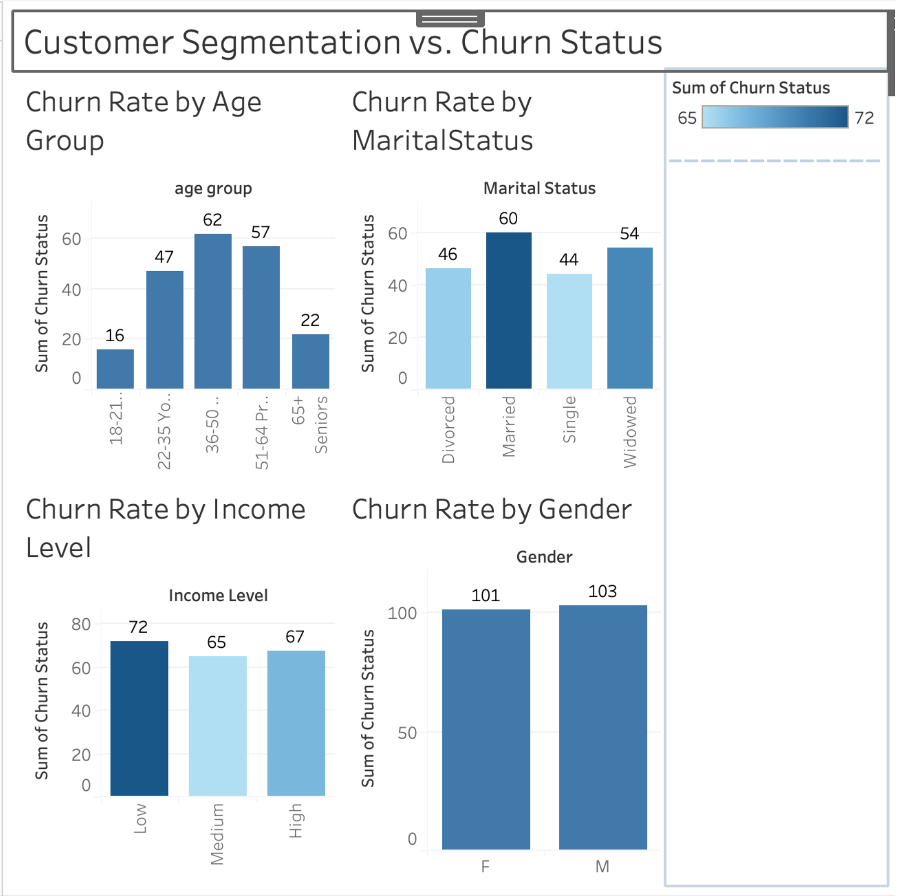
1. Tableau was used for visualisation to effectively explore patterns, relationships, and trends across multiple customer dimensions. The Customer Demographics worksheet provides a clear overview of customer segmentation and distribution, while behavioural worksheets — including transaction activity, digital platform usage, and customer service interactions — help assess customer engagement and experience. By analysing how these factors relate to churn status, the dashboards clearly highlight which aspects of behaviour and service experience have the most direct impact on customer churn rates.

Customer Demographics analysis

A customer segmentation dashboard was created in Tableau to provide a consolidated view of demographic distributions across age, gender, marital status, and income level. From the dashboard, it is evident that the majority of customers are concentrated within the 22–64 age range. In contrast, the distributions across gender, marital status, and income level appear relatively balanced, with no single group showing a dominant concentration.

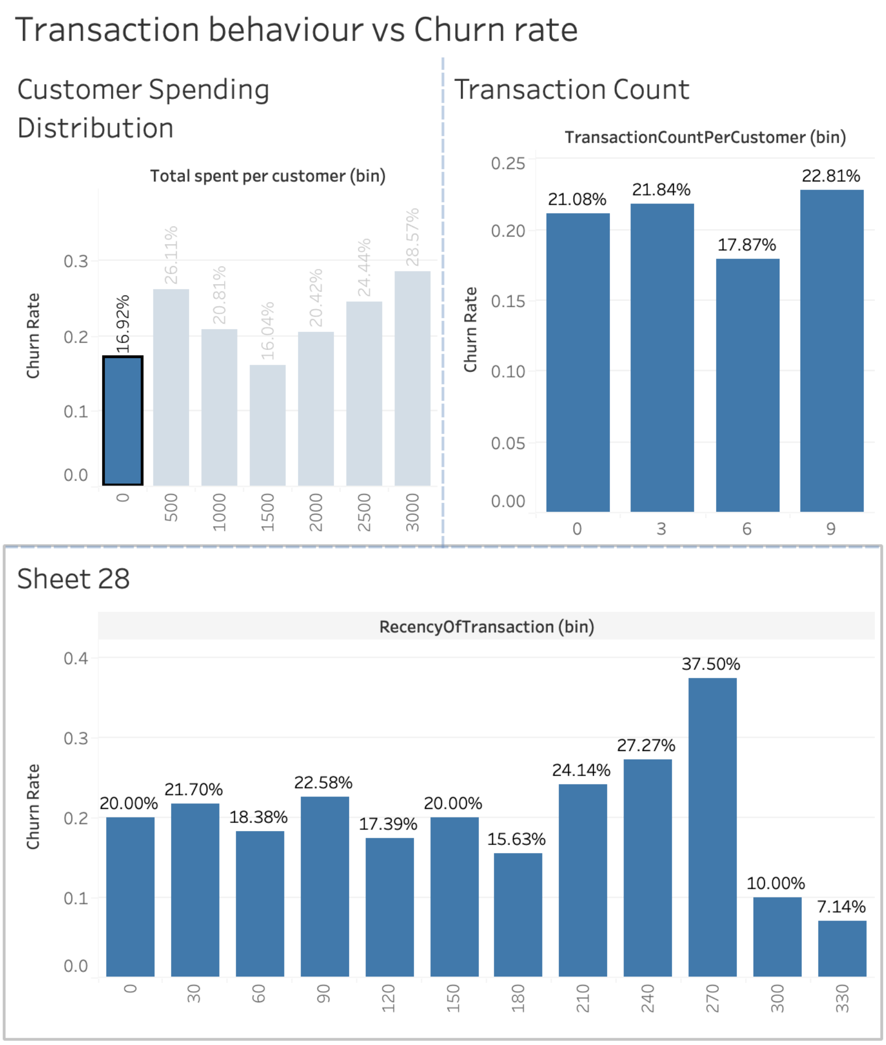


The customer segmentation vs. churn status dashboard clearly demonstrates how churn behaviour varies across different demographic groups, providing direct insight into which customer segments are more likely to leave. The analysis shows that churn is most pronounced among customers aged 36–50 and 51–64, while younger adults (18–21) and seniors (65+) exhibit comparatively lower churn levels. In terms of marital status, married and widowed customers display higher churn counts than single customers, suggesting life-stage factors may influence retention. Churn distribution across income levels appears relatively balanced, with a slightly higher churn observed among low-income customers. Gender-based churn shows minimal variation, indicating gender alone is not a strong driver of churn. These patterns highlight where targeted retention strategies should be prioritised, particularly for mid-age and higher-risk demographic segments.



Transaction Behaviour analysis

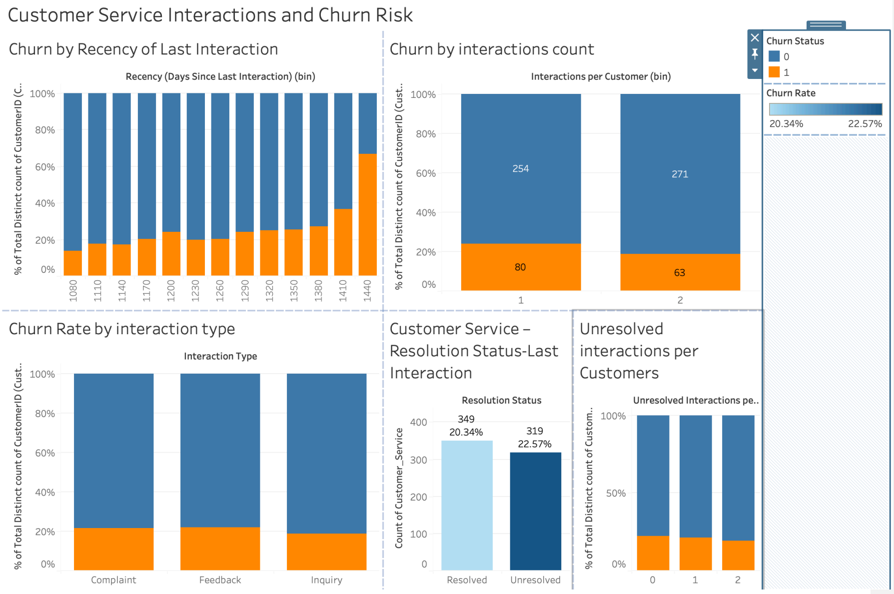
Overall, transaction recency is the most influential transaction-based predictor of churn, while total spend and transaction frequency show more complex and non-linear relationships. These findings suggest that monitoring inactivity and declining purchase recency is more effective for churn prevention than focusing solely on customer value or purchase volume.



Customer Service Interaction and Experience Analysis

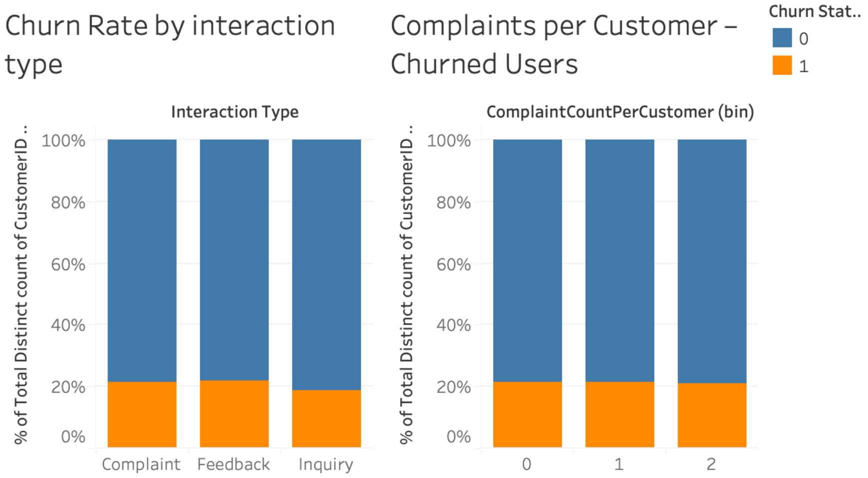
Commented [WW1]: How many customers had their most recent customer-service interaction in each month of 2022.

This analysis examines how customer service interactions influence churn behaviour, using interaction timing, frequency, type, and resolution status to identify early disengagement signals. The findings show that churn is driven less by customer contact volume alone and more by unresolved service experiences at the last interaction followed by inactivity.



Complaint-related interactions are associated with slightly higher churn rates compared to inquiries; however, the majority of customers who raise complaints do not immediately churn. This suggests that complaining is a form of continued engagement rather than an immediate exit signal. Customers often remain with the service after raising complaints, particularly when issues are resolved or followed up. Churn risk increases when complaints remain unresolved, occur repeatedly, or are followed by prolonged inactivity, indicating that it is the failure of service recovery rather than the act of complaining itself that drives customer attrition.

Impact of Customer Complaints on Churn

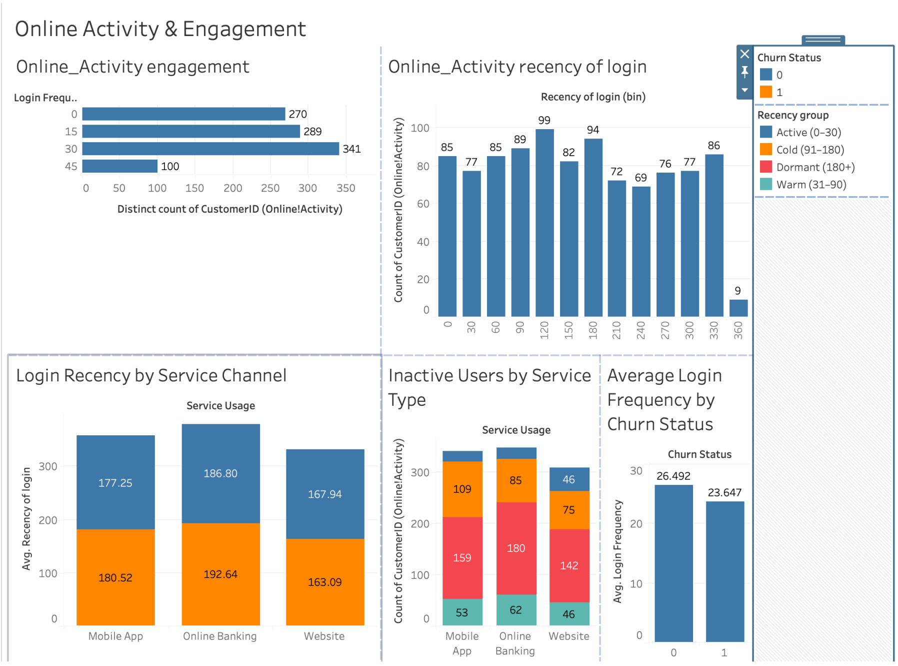


Complaint Resolution by Churn Status

Resolution ..	Churn Status 0	1
Resolved	124	32
Unresolv..	139	40

Online Engagement analysis

The analysis of online activity reveals a strong relationship between digital engagement and customer churn. Customers who exhibit lower login frequency and longer periods since their last login are more likely to churn, indicating that declining engagement serves as an early warning signal. Inactivity is particularly pronounced among online banking users, suggesting channel-specific disengagement risks. Overall, reduced digital interaction across login frequency, recency, and service channel usage is a key behavioural indicator of churn and highlights opportunities for targeted retention interventions.



Data Cleaning and Preprocessing Steps

The dataset required several cleaning and transformation steps before analysis. These steps ensured consistency across multiple tables (Customer_Demographics, Customer_Service, Transaction_History, Online_Activity and Churn_Status) and prepared the data for EDA and churn modelling.

A complete data-cleaning procedure was carried out using the pandas library in Python. The dataset was loaded into pandas using multi-sheet Excel handling, followed by an initial inspection using .info() and .head() to assess data structure, types, and overall quality. Duplicate records were identified and removed where necessary, and date and numerical fields were reviewed and standardised for accuracy and consistency.

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Missing values were examined and handled based on their business meaning. Categorical variables such as gender, marital status, and churn labels were cleaned and standardised, with additional validation checks performed to identify unexpected or inconsistent values. Feature engineering was then planned for analysis, including the creation of behavioural indicators such as login recency and total customer spend.

After completing all validation steps, the fully cleaned dataset was saved into new Excel output files. In practice, no data values were altered during the cleaning process. This is because the original dataset contained no missing, duplicated, or abnormal values. In addition, all feature engineering tasks were completed later within Tableau rather than in Python.

Outlier detection was conducted to identify values that could potentially distort descriptive analysis or downstream modelling results. Exploratory checks were performed using Microsoft Excel to review the distribution of key numerical and categorical fields, including customer age and customer identification counts. Summary statistics, sorting, and frequency checks were used to detect unusually high or low values and to confirm whether these observations reflected valid business cases or data quality issues.

Numerical feature standardisation or normalisation was considered as part of the data preparation process. As the analysis in this study focuses on exploratory data analysis and visualisation rather than predictive machine-learning modelling, features were retained in their original scales to preserve interpretability. Standardisation was therefore not applied at this stage. This step would be required if the dataset were later used for scale-sensitive machine-learning algorithms, such as logistic regression or clustering models.

For future machine-learning applications, categorical variables would require numerical encoding. Nominal features such as interaction type and service channel would be transformed using one-hot encoding to avoid introducing artificial ordering. Binary categorical variables such as resolution status would be encoded as 0/1 values. Identifier fields such as customer ID would be excluded from modelling, while derived numerical features including interaction frequency, complaint count, unresolved interactions, and recency would be retained in their numeric form.